
NSF TIP Directorate, X-Labs Outline: An Independent Research Organization for Physical AI Surgical Oncology

A Phase 1, first-in-human, LLM-advised eight-arm robotic Whipple with perioperative daraxonrasib (RMC-6236) in KRAS-mutated PDAC, by a performer that is neither a university nor a large company

Independent Research Organization

ChemicalQDevice

Mid-Scale

5 Years

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Submitted as an independent scientist under the funding approach set out in Science: A New Golden Age (July 2026). Not medical or regulatory advice, and not endorsed by the FDA, NIH, HHS, an IRB, ICH, NSF, or any sponsor. UC San Diego is named as the intended partner of choice and is not a sponsor, site, or endorser.

1 A Performer That Does Not Exist Yet

Science: A New Golden Age makes an argument about organizations, not only about money [1]. Federal agencies distribute roughly **\$200 billion** in annual research and development funding, and the report finds the portfolio defers to “the same incumbents that consume the funding.” Its Chapter II answers with a table scoring universities, corporate laboratories, federal laboratories, and new institutions against seven activities, and it names the NSF TIP Directorate’s X-Labs as “the first federal program designed to fund independent research organizations outside traditional academia.”

A first-in-human physical AI surgical oncology program is a clean case for that argument. A university department can run the trial but cannot maintain the engineering, the verification harness, and the regulatory package as a product. A device company can build the platform but will not publish negative operative results. A federal laboratory can verify the software but does not operate on patients. The work needs an organization that does all three and publishes, which is the definition report gives for a new institution in Table 1.

What the X-Labs model asks	What this organization is
A performer outside traditional academia	An independent company with no degree-granting affiliation and no indirect-cost negotiation
Mid-scale: tens of millions, ten to a hundred people, half a decade	A five-year program at the lower bound of that range, contracting one clinical site
Outputs that are public goods, not a single commercial asset	An open protocol, an open verification package, and a published safety set

Table 1. Three requirements the X-Labs model places on a performer, and the corresponding property of the organization proposed here. Each row is a claim the reviewer can check against the public repository.

2 Why No Existing Institution Fits

Chapter II’s institutional-fit table scores four performer types against seven activities. Applied to the five activities this program actually needs, the result shown in Figure 1 is the gap the X-Labs mechanism was created to fill: every existing type covers some, none covers all five, and the two that could operate on patients are the two least able to release the software openly.

d2-type / scored container grid

Activity this program needs	University dept.	Device company	Federal laboratory	This performer
Operate under an active IND	well suited	less suited	less suited	by contract
Maintain a verification harness	partly	well suited	well suited	well suited
Publish negative operative results	partly	less suited	partly	well suited
Hold regulatory package as a product	less suited	well suited	partly	well suited
Release code and data openly	partly	less suited	partly	well suited

Figure 1. The report's institutional-fit logic applied to the five activities this program needs. No existing performer type covers all five, and the clinical row is covered only by contracting a qualified site.

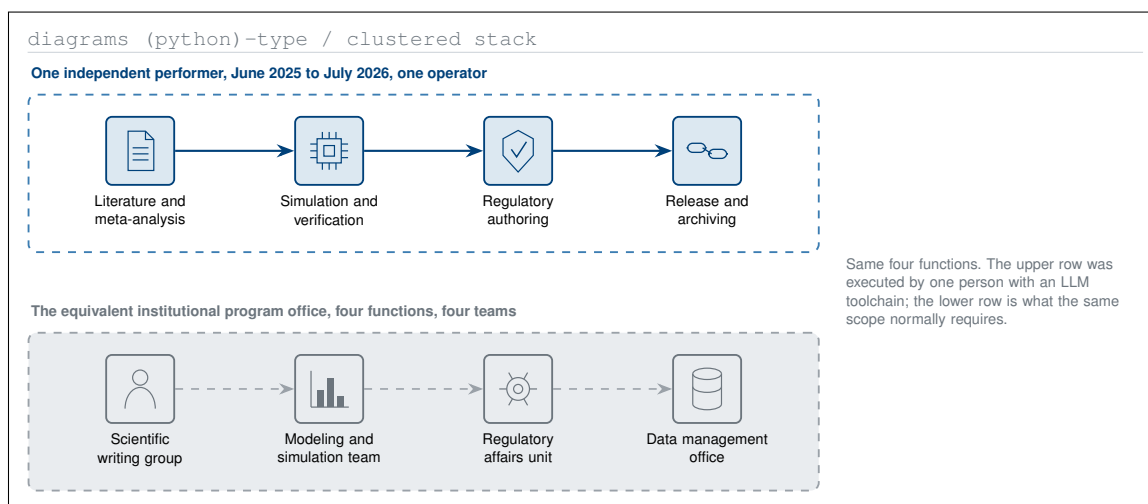


Figure 2. The same four functions, executed once by a single performer and once by the institutional program office the report describes. The comparison is the concrete form of the report's individual-scientist argument.

3 The Performer's Track Record

Between June 2025 and July 2026, without institutional support, one scientist identified the drug from forty pancreatic cancer meta-analyses, produced three quantitative trial simulations, and wrote a complete regulatory package as illustrated in Figure 2 [2, 3]. The August 2025 quantitative systems pharmacology run, ten arms and 250 ordinary differential equations per patient, returned median overall survival of **12.8 months** against **5.4** for chemotherapy [4]; RASolute 302 reported **13.2** against **6.6 months** nine months later [5]. The proximity is chronology, not validation: 1000 simulated against 241 enrolled, a combination against a single agent, KRAS G12C against a primarily G12D and G12V population highlighted in Table 2.

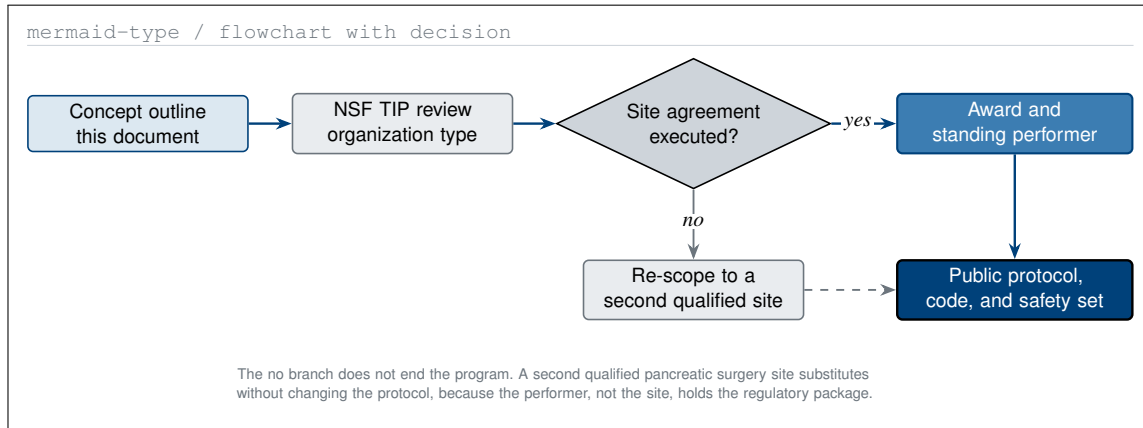


Figure 3. The pathway from this concept outline to a standing independent performer, including the branch taken if the first site agreement does not execute.

4 The Operation and the Advisory Boundary

The intervention is a staged eight-arm robotic pancreaticoduodenectomy with perioperative daraxonrasib and an on-premises language model whose output is advisory only [6, 7]. The model has no path to the arms: no generated token reaches a motion controller and no software in the advisory path holds write access to the command channel. Arm-level stop is specified at 3 milliseconds and system-wide stop at 500 milliseconds, measured on the bench before the first patient and again at every cohort boundary.

Parameter	Value	Source
Population	Resectable or borderline-resectable KRAS G12-mutated PDAC	Phase 1 protocol
Design and size	Open-label, single-arm, 3+3 escalation, up to 18 treated participants	Phase 1 protocol and IND
Co-primary endpoints	30-day device- or procedure-related SAEs; DLT, MTD or RP2D; tasks done with safe conversion	Phase 1 protocol
Follow-up	28-day screening; 30-day and 90-day safety; 24-month overall survival	Phase 1 protocol

Table 2. Trial parameters carried unchanged from the published Phase 1 protocol and the Investigational New Drug application, so concept is evaluated against a protocol that exists.

Positioning note. First in human refers to the integrated surgical and advisory workflow, not to daraxonrasib, which is investigational and already in Phase 3 evaluation. The robotic configuration is specified by manufacturer, model, cleared configuration, arm count, and software version at site agreement, and no drug supply or letter of authorization is in place with agent's developer.

5 Pathway, Budget, and the Partner Site

The request depicted in Figure 3 is **\$700,000 per year for five years**, \$3,500,000 total, with no cost share [8]. The organization is designed to be legible to the X-Labs review: a named performer, a contracted clinical site, public outputs, and a defined end state rather than an open-ended institute.

Moore's Cancer Center is the intended partner of choice, approached at the feasibility stage only: a nonconfidential scientific overview, a joint feasibility meeting with pancreatic surgery, gastrointestinal medical oncology, and trial operations, then an investigator-initiated intake with a budget and accrual estimate. UC San Diego is not a partner, sponsor, site, or endorser, because no written authorization exists.

Year	Public good released	Who can reuse it without permission
1	Bench stop-latency method and rig drawings	Any robotic surgery program running a comparable platform
2	Advisory logging schema and audit tooling	Any site running an on-premises model in a regulated workflow
3	Escalation dataset with negative results	Any sponsor designing a combined drug and device Phase 1
4 to 5	Full safety set, verification package, and archived code	Any investigator repeating or contesting the result

Table 3. The public goods released in each year of the term. The X-Labs case rests on these being usable by others without permission, which is what distinguishes the performer from a company laboratory.

Scope of Claims

Claimed (Table 3): that one independent performer produced, in fourteen months, the four functions an institutional program office normally divides across four teams, and that the outputs are public and citable. Not claimed: that the simulations validate RASolute 302, that the advisory system holds any regulatory clearance, that anything here has been used in a human, or that UC San Diego has agreed to anything. The trial does not exist until a site, IRB, an active IND, monitoring, insurance, and sponsor of record are in place.

Availability and Conflicts

Every work cited to a DOI is openly deposited; code, verification notebooks, and figure sources are in the repository [9]. The applicant holds no equity in Revolution Medicines or any robotic surgery vendor, and no sponsor, contract research organization, or medical society reviewed this document.

References

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